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MARKET MOVES PREDICTIONS USING RETRIEVAL-AUGMENTED GENERATION (RAG) ANALYSIS OF CAPITAL MARKET EXPERT OPINIONS IN SOCIAL MEDIA

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This study explores the predictive value of expert opinions from financial market media using Artificial Intelligence (AI), specifically, Retrieval-Augmented Generation (RAG) framework integrated with a Large Language Model (LLM). By analysing 3,877 YouTube videos spanning 12 months, the research categorised 4,808 expert opinions—either explicit or inferred—into directional market predictions (up, down, flat) for seven diversified financial assets. Results indicate that aggregated expert opinions correlate significantly with short-term market movements but lose predictive power for longer horizons. Explicit opinions demonstrated similar accuracy to inferred judgments, suggesting that LLMs effectively extract latent insights from unstructured data, enhancing accessibility and utility for retail investors. The study highlights the democratising potential of LLMs, providing timely and scalable analysis of vast datasets. However, challenges remain, such as understanding domain-specific nuances and speaker attribution within multimedia content. Statistical analysis reveals that expert opinions, particularly when aggregated, identify exploitable inefficiencies, thereby challenging the Efficient Market Hypothesis's assumption of perfect information dissemination. Short-term market anomalies observed align with behavioural finance theories of cognitive bias and delayed information diffusion. By bridging qualitative sentiment with quantitative modelling, this research underscores the transformative role of AI-driven tools in financial analysis, offering new avenues for individual and institutional investors. Further development of LLMs tailored to domain-specific complexities may revolutionise investment practices and advance research on market behaviour.

Keywords: AI; RAG; LLM; market movement prediction; investment; social media; financial markets

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1. Introduction

Financial markets are increasingly shaped by the dynamics of expert commentary disseminated via digital platforms, particularly social media and video-based channels. Retail investors, in particular, rely on these expert opinions to form judgments under uncertainty, often perceiving them as more timely and accessible than institutional reports (Chaitanya & Nordin, 2021; Gaikwad et al., 2023). The proliferation of financial commentary across platforms like YouTube has significantly increased the visibility of analysts and independent experts, transforming the traditional flow of information and its role in market reactions (Vamossy, 2024).

Previous studies have confirmed that sentiment extracted from financial texts, such as news headlines or earnings reports, can predict short-term market fluctuations, suggesting a measurable influence of media narratives on investor behaviour (Tetlock, 2007; Blasco, Corredor & Ferreruela, 2012). This predictive potential is grounded in behavioural finance, which highlights cognitive biases, social influence, and information cascades as mechanisms that cause market participants to respond in patterned, often irrational ways (Barberis, Shleifer & Vishny, 1998; Shiller, 2003). LLMs, particularly when integrated into RAG frameworks, offer new methodological possibilities for capturing these latent expert sentiments. Unlike traditional sentiment tools, LLMs can infer meaning from context-rich, unstructured data such as video transcripts, where expert judgments may be implied rather than explicitly stated (Gao et al., 2024; Zhao et al., 2024). This capability extends beyond text classification, enabling semantic extraction, contextual interpretation, and scalable analysis—making such tools increasingly relevant in real-time financial analytics (Delgadillo, Kinyua & Mutigwe, 2024). This study investigates whether expert opinions retrieved from YouTube financial channels—either directly stated or inferred using a custom RAG-LLM pipeline—can predict short-term movements in key financial assets.

2. Literature Review and Research Gap

Expert opinions disseminated via financial news media and social platforms significantly influence investor behaviour, particularly among retail investors. These platforms act as informal channels of financial education and information sharing, shaping perceptions of market trends and risks (Chaitanya & Nordin, 2021). Research indicates that retail investors perceive social media content as timely and, in many cases, more trustworthy than formal financial reports, particularly when authored by influential analysts or known professionals (Gaikwad et al., 2023). These dynamics are intensified by emotional contagion and group reinforcement mechanisms common to digital platforms (Vamossy, 2024). Seminal work by Tetlock (2007) showed that textual sentiment in financial news correlates with short-term stock returns, laying the foundation for computational sentiment analysis in finance. Similarly, Blasco et al. (2012) demonstrated that perceived expert consensus often drives market imitative behaviour.

Furthermore, previous work on crowd-sourced or panel-based expert forecasting indicates that diversity of opinion can improve predictive performance, particularly when aggregating across heterogeneous sources (Chandra & Kumar, 2011; Baltakienė, Kanniainen & Baltakys, 2021; Bender et al., 2022). Such findings align with behavioural finance, emphasising bounded rationality, overreaction, and delayed information assimilation in driving asset price fluctuations (Barberis, Shleifer & Vishny, 1998; Shiller, 2003). The Efficient Market Hypothesis (Fama, 1970) postulates that all publicly available information is reflected in prices. However, short-term inefficiencies — such as post-earnings drift or delayed reaction to the news — have been documented (Jegadeesh & Titman, 1993; Barberis, Shleifer & Vishny, 1998). These transient inefficiencies may be exploitable by real-time sentiment or expert opinion aggregation, especially using advanced AI tools.

Despite advances in natural language processing, most sentiment analysis applications in finance remain confined to structured textual data such as headlines, tweets, or earnings calls (Gu et al., 2024; Heydarian, Bifet, & Corbet, 2025). The unstructured and conversational nature of long-form video interview content, which is similar to earnings conference calls —where judgments may be implicit or context-dependent —poses a greater analytical challenge (Cao et al., 2024). Conventional models like FinBERT or VADER are ill-equipped to infer nuanced sentiment from such data (Yang, UY & Huang, 2020). Recent developments in LLMs, particularly when embedded in RAG systems, enable deeper contextual understanding and knowledge-grounded reasoning (Gao et al., 2024; Yu et al., 2024). These models can retrieve, interpret, and synthesise expert opinions from extensive multimedia datasets, offering new possibilities for market signal extraction (Delgadillo, Kinyua & Mutigwe, 2024).

Despite growing interest in AI-driven financial analysis, three major gaps persist: first, few studies focus on qualitative expert opinion in unstructured video formats, despite their popularity and influence; second, LLMs and RAG systems have not been extensively tested for inferential prediction based on expert sentiment — going beyond merely stated forecasts to interpret implicit views; third, there is limited empirical evidence on whether these LLM-extracted insights outperform or complement explicit expert predictions in the context of real-time market behaviour. This represents a significant omission, as LLMs may uncover latent sentiment and enhance the accessibility of advanced financial analysis for non-professionals. This study addresses this gap by applying

a custom RAG-LLM pipeline to extract and classify 4,808 expert opinions from financial YouTube videos. It tests their alignment with short-term asset price movements across seven diversified instruments. The research contributes to behavioural finance by quantifying expert opinion accuracy and evaluating the predictive validity of inferred versus direct sentiment across a multimedia corpus.

3. Theoretical background

EMH assumes that financial markets fully reflect all available information, making it impossible to consistently achieve returns that outperform the market through analysis or prediction (Fama, 1970). While the EMH is seen as a cornerstone of modern financial theory, its universal applicability has been questioned, particularly in light of reported market anomalies and the evolving role of technological tools in financial analysis. The anomalies contradicting the EMH include, for example, the momentum effect, the calendar, such as "January" effects, or value-growth disparities (Fama & French, 1992; Jegadeesh & Titman, 1993; Cheema, Ding & Wang, 2023). The persistence of specific inefficiencies, such as overreaction and underreaction to earnings announcements (De Bondt & Thaler, 1985) and price drifts following news events (Barberis, Shleifer & Vishny, 1998) and identifying psychological biases and irrational behaviour as key drivers of short-term inefficiencies (Shiller, 2003) was also reported. These anomalies may suggest systematic departures from efficiency, creating opportunities for investors to exploit (Jegadeesh & Titman, 1993). The Grossman-Stiglitz Paradox (Grossman & Stiglitz, 1980) asserts that markets cannot be perfectly efficient, as the incentive to gather costly information diminishes in such a state. Active portfolio management leveraging segmented market information has been shown to yield superior results under specific conditions (Kacperczyk, Sialm & Zheng, 2005; Zhou et al., 2025), further contradicting key assumptions of the EMH.

4. Research objective and methodology

The integration of LLMs and RAG systems into financial opinions analysis introduces a new dimension to the process, supporting the processing of large amounts of unstructured data such as expert opinions. Tetlock (2007) demonstrated the predictive value of textual sentiment analysis, which aligns with the enhanced capabilities of LLMs in uncovering actionable signals. These tools challenge the EMH's implicit assumption of equal accessibility and interpretability of public information by extracting latent insights from qualitative data (Zhao et al., 2024).

This research aims to explore whether expert opinions from media sources are predictive of subsequent market movements. The study seeks to answer the two research questions: (1) Are diverse expert opinions pulled from recordings of industry-paid professionals and independent financial journalists predictive of actual market movements? (2) Are explicit opinions, directly stated by experts, more predictive of asset price movements than inferential judgments made by LLMs based on the broader context of the discussions and their internal knowledge? By comparing explicit expert opinions expressed in the YouTube (YT) videos with inferential judgments made by LLM, the study aims to assess whether the latter provides additional predictive value beyond direct expert commentary and whether LLM can be used to enhance the interpretation of financial market opinions when direct signals are not readily available.

In this study, qualitative predictions of the market movements expressed in YT videos were converted into 3 categorical variables reflecting the direction of the prediction (up, down or flat). Statistical analysis was employed to assess the association between the predicted and actual market movements. The BLINDED LLM-based tool developed by the research team to support evidence synthesis and analysis was used to support the analysis of the video scripts and assign one of the pre-defined categories. The accuracy of the analysis was manually confirmed on 10% of the analysed sources, demonstrating a high level of accuracy. Any discrepancies in the judgment were resolved in a discussion between DG, AB and MG.

To address these questions 3877 YouTube (YT) videos covering 7 financial assets (tickers) over the 12 months from October 2023 to September 2024 were analysed. A database of 17953 YouTube videos as the primary data source was assembled (Those video transcripts were downloaded using YouTube API, integrated into BLINDED tool. At the dates of uploading minor number of videos did not have the text transcripts. We

considered this fact as insignificant for the size and quality of final dataset). 78 relevant channels were selected, discussing the capital and financial market issues. This selection ensured representation of the mainstream financial media (Bloomberg, Yahoo Finance, CNBC, etc.), smaller organisations, independent experts, and financial journalists. The former sources are assumed to host highly ranked financial experts, and the latter cater to niche audiences, potentially offering alternative or contrarian viewpoints to the consensus often found in mainstream coverage. Selected videos feature a diverse range of financial experts discussing market trends and predictions for Gold (Gold, or Gold CFD (ticker XAUUSD), or Gold ETFs (ticker GLD)), Bitcoin (Bitcoin (ticker BTCUSD) or Bitcoin ETFs), the S&P 500 (S&P500 index (ticker SPX) or S&P500 ETF (ticker SPY) US Treasury Bonds, Crude Oil, Tesla Inc. and Alphabet Inc. Some of the sources contained multiple opinions. LLM was allowed to identify no more than 2 opinions per source, resulting in 4808 opinions analysed (see Table 1). The correctness of this extraction was verified by the extra prompt aimed at detecting the number of experts.

Table 1. Breakdown of the opinions included in the analysis.

Asset	Number of YT videos	Number of Opinions	Direct Opinions	Inferred Opinions
Gold	724	704	411	293
Bitcoin	738	826	333	493
S&P 500	883	986	349	637
US Treasuries	392	383	152	231
Oil	685	618	299	319
Tesla Inc.	830	841	217	624
Alphabet Inc.	487	450	90	360
	4739*	4808	1851	2957

Note. The total number of YT videos was 3877, which is less than the sum of the second column because some videos contained recommendations for several tickers.

The assets selection criteria included: 1) diversification; 2) reasonable coverage in the selected YT videos; 3) high level of capitalisation (assets under management) and trading volumes. Research supports the assumption that portfolio diversification reduces idiosyncratic risks and enhances stability by spreading investments across different asset classes and regions, as it shows that diversification can mitigate specific asset risks and improve portfolio stability (Upson, Jessup & Matsumoto, 1975). Studies also indicate that holding a variety of assets can offset losses in one area with gains in another, thereby reducing overall portfolio volatility (Zaimovic, Omanovic & Arnaut-Berilo, 2021). This strategy is effective in achieving a more stable return profile (Woerheide & Persson, 1992). Goetzmann and Kumar, 2001 found that diversification reduces idiosyncratic risk, which is crucial during volatile periods. Furthermore, Clare et al. (2016) highlight that incorporating various asset classes, such as bonds, can improve risk-adjusted returns. A well-diversified portfolio can improve risk-adjusted returns, particularly for retail investors who may lack extensive resources for asset analysis, reducing unsystematic risk specific to individual investments (Zaimovic, Omanovic & Arnaut-Berilo, 2021).

Literature supports the importance of measuring asset correlation for diversification effectiveness. Low or negative correlations between assets can enhance diversification by ensuring that assets respond differently to market changes (Koumou, 2020). However, factors such as transaction costs, limited access to diverse assets, and biases like home bias can hinder true diversification (Nguyen et al., 2023). These impediments can lead to suboptimal portfolio construction, reducing the potential benefits of diversification (Zaimovic, Omanovic & Arnaut-Berilo, 2021). Assets selected for this study have low or negative correlations, as shown in Table 2. Moreover, all correlations are insignificant, except for TLT and USO, which is negative.

Table 2. Correlation matrix for selected assets (tickers). Monthly returns.

		TLT return	GOOG return	TSLA return	GLD return	GBTC return	USO return	SPY return
TLT return	Pearson's r	—						
	df	—						
	p-value	—						
GOOG return	Pearson's r	-.141	—					
	df	11	—					
	p-value	.645	—					
TSLA return	Pearson's r	.192	.165	—				
	df	11	11	—				
	p-value	.529	.591	—				
GLD return	Pearson's r	-.045	.036	-.243	—			
	df	11	11	11	—			
	p-value	.883	.907	.424	—			
GBTC return	Pearson's r	-.085	-.092	-.150	.201	—		
	df	11	11	11	11	—		
	p-value	.783	.766	.625	.510	—		
USO return	Pearson's r	-.587*	.347	.090	-.300	.085	—	
	df	11	11	11	11	11	—	
	p-value	.035	.246	.771	.320	.783	—	
SPY return	Pearson's r	.447	.225	.296	-.140	.437	.185	—
	df	11	11	11	11	11	11	—
	p-value	.126	.459	.326	.649	.135	.546	—

Note. Correlation tests performed for the monthly returns computed as closing prices for the first day of the respective months divided by the respective value of the previous month. Source of data: <https://www.tradingview.com/>. Jamovi 2.3.28 was used for computations.

* $p < .05$

The analytical approach centres on using a RAG system empowered by an LLM (described below) to extract and synthesise expert opinions from the transcripts of these videos. The opinions extracted were categorised into either explicit or inferential. Explicit opinions are those directly stated by the experts in the recordings, reflecting a clear expectation of price movement. In cases where no explicit opinion was presented, the LLM drew inferential conclusions from the text, identifying the likely sentiment based on the broader context of the discussion.

RAG is a framework designed to enhance the capabilities of language models by integrating external knowledge sources into the generation process (Yu, 2022; Yu et al., 2024). The primary purpose of RAG is to improve the accuracy and relevance of generated content by retrieving pertinent information from a vast corpus of documents, which is then used to inform the generation process (Gao et al., 2024; P. Zhao et al., 2024). This approach allows RAG to overcome the limitations of traditional language models that rely solely on pre-trained data, which may not be up-to-date or comprehensive (Yu, 2022; Gao et al., 2024). The RAG framework consists of two main components: a document retriever and a text generator (Zhao et al., 2024; Yu et al., 2024). The document retriever is responsible for identifying and retrieving relevant documents from a large dataset based on the input query (Yu, 2022; Gao et al., 2024). RAG's ability to incorporate external knowledge dynamically makes it particularly useful for tasks that require up-to-date information, such as question answering and fact verification (Gao et al., 2024; P. Zhao et al., 2024). By leveraging both structured and unstructured data sources, RAG can provide more accurate and diverse responses compared to models that rely solely on pre-trained data (Gao et al., 2024; Yu et al., 2024). This integration of retrieval and generation processes not only enhances the model's performance but also improves its interpretability, as the generated content can be traced back to specific data sources (Zhao et al., 2024; Yu et al., 2024).

A RAG approach was implemented in three steps. First, textual content (transcripts for videos) was converted into dense embeddings using the 'text-embedding-3-large' model (<https://openai.com/index/new-embedding-models-and-api-updates/>). The 'text-embedding-3-large' model is specifically designed to create high-dimensional vector representations of text, encapsulating semantic meaning. This capability is crucial as it allows for efficiently transforming text into embeddings, facilitating various natural language processing tasks

(Petukhova, Matos-Carvalho & Fachada, 2024). By leveraging LLMs, these embeddings enhance the semantic understanding of text. They achieve this by capturing the context and relationships between words. (Cao, 2024). The efficiency in transforming text data into embeddings is a significant advantage of using LLMs. This efficiency is realised through the model's ability to process large corpora and generate embeddings that reflect a comprehensive understanding of language variations (Harris, Butani & Hashmy, 2024). In the second step the similarity search was conducted using cosine similarity algorithm (<https://www.pinecone.io/learn/vector-similarity/>) offered by Pinecone.io. This step involved applying the user's query to the text chunks identified as most relevant in the previous stage. For this task, the ChatGPT-4o-mini was used. In the third step, the ChatGPT-4o model was used to extract and analyse information from the selected text chunks, utilising its advanced capabilities in language understanding and generation.

The exponential moving average (EMA) is an effective method for measuring medium-term tendencies in stock price analysis due to its ability to give more weight to recent prices, allowing it to respond more quickly to price changes than a simple moving average (Lorig, Zhou & Zou, 2019; Billah et al., 2024). It makes it suitable for capturing medium-term trends in stock prices, as it can quickly adjust to changes in market conditions (Droke, 2001; Arisoma & Narulita, 2019). The EMA is often used in financial markets to smooth out price data and identify trends, making it a valuable tool for validating expert predictions (Droke, 2001; Raudys & Pabarškaitė, 2018).

For assets included in the analysis, EMA can help validate predictions by highlighting the underlying trend and filtering out short-term volatility (Droke, 2001; Arisoma & Narulita, 2019). The EMA's responsiveness to recent price changes makes it particularly useful in volatile markets, where quick adjustments to new information are crucial (Raudys & Pabarškaitė, 2018; Arisoma & Narulita, 2019). Applying the EMA to these assets makes it possible to gain insights into the medium-term direction of prices, aiding in the assessment of expert forecasts (Droke, 2001; Raudys & Pabarškaitė, 2018). Our research used 4 weeks EMA (All indicators were calculated using www.tradingview.com).

The ATR is a well-established measure of market volatility, introduced in "New Concepts in Technical Trading Systems" (Wilder, 1978). It is widely used as a volatility indicator in technical analysis and trading systems (Edwards, Magee & Bassetti, 2018). The ATR's ability to measure volatility makes it a versatile tool in both long and short-trading systems (Gayed, 2015). It is now used in various trading systems and strategies, including trend-following systems like the parabolic stop and reverse (SAR) (Yamanaka, 2012). Additionally, ATR is used in conjunction with other indicators to improve trading systems. The ATR's effectiveness in capturing market volatility is supported by its inclusion in numerous studies and its application in different market conditions (Edwards, Magee & Bassetti, 2018). The widespread use of ATR in various trading systems and its inclusion in technical analysis literature further support its acceptance in the finance community.

We are using a fraction of ATR to establish if the underlying asset's month-over-month price movement is different from the typical volatility for this asset. The selection of a fractional threshold should support the purpose of research and is heavily affecting the hit rate. We opted for 5% (500 basic points) intervals across the range spanning from 5% to 50% threshold. For each of the 7 assets the 4 months ATRs were calculated (All indicators were calculated using www.tradingview.com), followed by applying 12 months simple moving average to arrive to a single smoothed ATR value per asset.

Extracted opinions are assigned to one of two groups as assessed by the LLM (direct or inferred). Each opinion was categorised as either Up, Flat or Down, reflecting the experts' prediction of the expected price movements in the next few weeks. The opinions were counted for each month, and a weighted average for each month was calculated using the formula:

$$(Up-Down) / (Up + Flat + Down)$$

to assess the general price movement prediction for each asset in each month. If the module of weighted score was smaller than .1 then the price movement was categorised as 'Flat' and assigned 0. The 'Up' price movement was assigned 1 when the weighted score was larger or equal to .1, and 'Down' price movement was assigned -1 if the weighted score was less or equal to -.1.

These values were then compared to factual price dynamics by calculating the annualised monthly returns. For the month of opinions observation (current month), this month's last week 4-weeks exponential moving price average (4W-EMA) was divided by the respective value of the previous month, minus 1, and multiplied by 12 for the convenience of the price dynamics perception (current month annualised return). For the month following the current month (next month), the next month's last week 4W-EMA was divided by the respective value of the current month, minus 1, and multiplied by 12 (next month annualised return).

In addition, the month-to-month EMA change values were transformed into the categorical price move assessment by comparison with various thresholds established as fractions of ATR: -1 for negative price moves, 0 for sideways dynamics, and +1 for positive moves. The rationale for using different ATR fractions is that assigning a 'flat' label to a certain range of price volatility is a discretionary decision influenced by the purpose of analysis, target returns and risk profile.

Finally, 'Hit' scores reflected the comparison result, where 1 was assigned to matching values and 0 to mismatches, considering matching of 'flat' prediction with a sideways market move computed using ATR fraction values sampled at 5% intervals across the range spanning from 5% to 50%, and "up" and "down" by matching with positive or negative market move beyond the sideways range in each ATR threshold case.

The Hit Rates were calculated to assess the proportion of correct predictions in each of the analysed groups. Different thresholds were applied for the flat trend in the assets included (see Table 3). The correlation analysis between weighted scores with the current and next month's returns was used to see whether the predictions and actual market movements were associated. Wilcoxon signed-rank test was employed to explore differences between the accuracy of direct and inferred opinions.

5. Results and discussion

Considering a 0.5 hit rate as indicating a 50/50 chance of predicting the market movement, applying narrower volatility ranges for the sideways dynamics returns better hit-rates. Table 3 represents average hit-rates for all seven tickers, and the ticker-by-ticker grouping is provided in Appendix A. While 'flat' thresholds from 5% to 30%, applied to the current market move, return chances to predict exceeding 50/50, the same approach for next market move does not show such accuracy. In the best option - 5% ATR - only one direct prediction (USO) and three inferred predictions (TLT, USO and GOOG) for the current month's end-of-month (EoM) EMA meet the .5 threshold, while the remaining values exceed it. Meanwhile, the combined total hit rate (direct plus inferred) for each ticker falls within the range [0.67, 0.83], indicating that the chances of the blended expert panel predicting the market movement are better than 50/50, if the sideways dynamics is defined as equal or less than 5% of ATR. In contrast, the same combined measure for the market movement of next month's EoM EMA within this ATR range is below 0.5 for four out of seven assets, indicating that the expert panel fails to predict the market movement over the longer-term horizon.

Both explicit and inferred expert opinions correlated significantly with current-month price movements ($p < 0.001$), demonstrating their ability to predict market trends within the same month. Diversity of perspectives seems to enhance the reliability of predictions over shorter timeframes by incorporating varied methodologies and viewpoints. However, the predictive power of these opinions does not extend to longer-term horizons, as no significant correlations with next month's price movements were observed.

Table 3. Average hit-rates for seven tickers

Fraction of ATR assigned for 'Flat'	Average hit rate (opinions are equal to current EoM price move) direct	Average hit rate (opinions are equal to current EoM price move) inferred	Average hit rate (opinions are equal to current EoM price move) total	Average hit rate (opinions are equal to next EoM price move) direct	Average hit rate (opinions are equal to next EoM price move) inferred	Average hit rate (opinions are equal to next EoM price move) total
5.00%	0.67	0.63	0.71	0.51	0.5	0.51
10.00%	0.62	0.6	0.69	0.46	0.5	0.49
15.00%	0.6	0.6	0.67	0.45	0.49	0.48
20.00%	0.58	0.58	0.63	0.4	0.44	0.46
25.00%	0.55	0.58	0.61	0.38	0.45	0.45
30.00%	0.54	0.51	0.56	0.37	0.46	0.43
35.00%	0.54	0.5	0.55	0.36	0.48	0.44
40.00%	0.54	0.49	0.52	0.35	0.44	0.4
45.00%	0.46	0.48	0.48	0.36	0.43	0.38
50.00%	0.45	0.44	0.44	0.31	0.4	0.33

All the correlations of weighted scores (either direct or inferred) with the current month's return were statistically significant ($p < .001$), which shows the positive significant association between the expert panel opinion and actual market moves for the short-term perspective (see Table 3). Moreover, direct ($M = 0.281$, $SD = 0.495$) and inferred ($M = 0.292$, $SD = 0.379$) weighted scores also correlate with each other ($r = .576$, $p < .001$), which indicates that LLM's inference on expected market moves derived from the conversations and wider context may be associated with the explicitly expressed expert opinions.

To test the difference between the direct and inferred opinions, a paired-sample t-test was conducted. The data were normally distributed, as assessed by the Shapiro-Wilk test ($W = 0.974$, $p = 0.088$). The results indicated no significant difference between the two groups, $t(83) = -0.247$, $p = 0.805$, $d = -0.027$. This finding answers the second research question: For the current months' time horizon, explicitly stated expert opinions do not differ from inferences made by the LLM from non-explicit recommendations. None of the correlations of weighted scores (whether direct or inferred) with the next month's return was statistically significant. Thus, the data does not provide enough evidence to support an association between weighted opinion scores and market movement on a time horizon longer than a month.

Explicit opinions were found to be no more predictive than inferential judgments. Both types of predictions demonstrated similar levels of correlation with current-month price movements, indicating that LLMs can effectively infer market expectations from broader contextual cues and implicit sentiments in the discussions.

Both explicit and inferred predictions were found to be associated with the market movement in the short term (within the same month the predictions were made) and performed similarly in terms of their predictive accuracy. The longer-term predictive accuracy was not confirmed. These findings suggest that the diversified set of opinions shared through YT channels may provide support for investors in making short-term decisions with a large volume of information needed to be analysed in a very short time. LLM based tools such as BLINDED tool, can offer this ability, offering speed of processing data exceeding ability of human analyst and providing quick 'as you need' support for individual investors.

The similar accuracy of direct and inferred opinions shows the potential of LLM based tools to interpret and summarise a wide range of non-direct data. This could potentially offer a high-level specialist opinion to different groups of investors, improving the accessibility of the industry (Yang, Tang & Tam, 2023). In the longer term this could lead to a change in market dynamics resulting from democratisation of investment practices (H. Zhao et al., 2024).

Using the RAG framework provided an effective solution for extracting and synthesising expert opinions from video transcripts in the financial markets' domain. By combining the document retrieval capabilities of RAG with an LLM, we could efficiently capture nuanced opinions on the selected asset price movements, producing

robust results. RAG's source selection process generally performed well, consistently retrieving relevant content and enabling effective data extraction.

This research also highlighted a few challenges in using an LLM based tool in the analysis process. The BLINDED tool dealt well with information related to all types of analysed assets, however the TLT analysis required a change in the prompt due to the inverse character of prices and yields for fixed income instruments: when the yield goes higher ('long' movement as the price rise for equities), the prices of bonds go down. Lack of understanding of this directional relationship initially led to some inaccuracies in interpreting recommendations. This demonstrates the need for careful formulation of prompts, fitting with the characteristics of the specific data, rather than using 'one fit all' approach (Krause, 2023). This may be particularly important when LLM based tools are introduced to the users, especially those less familiar with this technology (Zamfirescu-Pereira et al., 2023). Especially as an increasing amount of research highlights the multiple misconceptions users may have regarding the capabilities of LLMs, expecting them to 'think' and 'know things' (Bowman, 2023; Birhane & McGann, 2024; Wang et al., 2024). Additionally, the LLM occasionally failed to distinguish between different speakers within a single video, causing minor issues in attributing expert opinions accurately. These limitations underscore the importance of refining LLM capabilities, especially in areas requiring a clear understanding of financial instruments and speaker identification within complex multimedia sources.

The comparable predictive accuracy of inferred and explicitly stated expert opinions demonstrates the utility of latent information embedded in textual data. This finding aligns with prior research emphasising the value of sentiment analysis in financial contexts. For instance, Tetlock (2007) demonstrated that qualitative analysis of financial texts could predict stock price movements by capturing underlying sentiment and market expectations. Inferred insights reflect a combination of explicit signals and contextual nuances thus potentially enhancing traditional analysis (Gu et al., 2024). This underlines the importance of advanced analytical tools in extracting actionable information from large, unstructured datasets.

The predictive power of aggregated opinions may also complement existing factor-based models of asset pricing. For instance, the Five-Factor Model accounts for much of the variation in returns between diversified portfolios by integrating independent risk factors (Fama & French, 2015). However, the ability of inferred opinions to predict price movements suggests that latent information, not explicitly modelled, may act as a supplementary risk factor. This finding encourages further research into the intersection of qualitative sentiment analysis and quantitative risk modelling.

In the study, we examined the predictive capacity of expert opinions within the framework of market efficiency. Our findings indicate that aggregated expert opinions, particularly when interpreted through advanced tools like LLMs, can uncover exploitable inefficiencies. This suggests that while aggregated expert opinions exhibit short-term predictive power, their accuracy diminishes over longer horizons. While the long-term efficiency of markets is supported, our findings reveal statistically significant short-term predictive power in expert opinions. These results suggest that short-term inefficiencies exist, likely arising from delays in the dissemination or interpretation of information by market participants (Jegadeesh & Titman, 1993).

This supports a dynamic interpretation of EMH, wherein market efficiency is temporally bounded and contingent upon the speed of information dissemination and integration (Chordia, Roll & Subrahmanyam, 2005). The theory of delayed information diffusion further explains such anomalies, positing that inefficiencies arise temporarily as markets adjust to new data (Hong & Stein, 1999). Behavioural finance literature provides another explanation for such inefficiencies, highlighting cognitive biases, such as overreaction or underreaction to news, as drivers of these transient anomalies (Barberis, Shleifer & Vishny, 1998). These biases create temporary windows of opportunity where informed actors or advanced analytical systems can exploit market dislocations before prices fully adjust.

The findings also intersect with the Joint Hypothesis Theorem, which highlights the inseparability of tests of market efficiency from the models of market equilibrium employed (Fama, 1970). Observed short-term

inefficiencies could result from limitations in existing equilibrium models rather than a failure of market efficiency itself. For example, the Fama-French Five-Factor Model extended the traditional Capital Asset Pricing Model (CAPM) by incorporating factors such as size, value, profitability, and investment intensity (Fama & French, 2015). However, new factors—such as the rapid dissemination of information via social media or the influence of algorithmic trading—may represent emerging dimensions of market behaviour that are not fully captured by existing models.

A selection of YT videos published in English was selected for analysis using the available resources. Although the sample characteristics seem to reflect the opinions on the topic well in the selected timeframe, the narrower selection could introduce bias or skew data in one direction or another. For practical reasons, seven different assets were selected for analysis. This limits the conclusions to these assets only.

Conclusions

This study demonstrates that aggregated expert opinions extracted from YouTube media using LLM-enhanced RAG frameworks can predict short-term market movements significantly. The novel contribution lies in showing that inferred opinions generated by LLMs from unstructured expert discourse are just as predictive as explicitly stated ones—an insight not previously documented in the financial AI literature. This expands the analytical toolkit available for interpreting informal financial commentary and challenges the assumption that only formal, explicitly declared views are of value.

The research contributes to behavioural finance and AI-driven market forecasting by evidencing that aggregated media sentiment—when processed via state-of-the-art neural models—captures transient inefficiencies in asset pricing. These inefficiencies are short-lived and suggest that market participants do not always incorporate qualitative insights immediately, providing a potential edge for retail investors using LLM tools.

There are several limitations. The dataset is restricted to English-language YouTube videos and seven financial assets, limiting the generalisability of the results. Attribution errors in speaker identification and the inverse nature of bond yield-price relationships posed challenges for LLM interpretation. Furthermore, the study's predictive power diminishes over longer horizons, indicating that while LLMs can summarise market mood effectively, they cannot fully anticipate delayed reactions or structural shifts.

The results underscore the potential of generative AI in synthesising decentralised and noisy expert discourse, offering a scalable means to support investor decision-making in the face of overwhelming information volumes.

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